

Compilation Time Benchmark: Clyte, GCC, and Clang

In this benchmarking study, we compare the compilation times of three compilers: Clyte, GCC, and Clang. The comparison involves three programs of different sizes, from small to large. Each program is compiled 5 times to obtain an average compilation time, ensuring fair results by clearing caches between runs.

Small Program Test: small.c

Program Code:

```
int main()
{
    int sum = 0;
    int i = 0;
    for (i = 0; i < 1000; i = i + 1)
        sum = sum + i;
    return sum;
}
```

Clyte Compilation Times

Run	Real Time	User Time	Sys Time
1	0m 0.019s	0m 0.002s	0m 0.000s
2	0m 0.018s	0m 0.000s	0m 0.001s
3	0m 0.017s	0m 0.001s	0m 0.000s
4	0m 0.019s	0m 0.001s	0m 0.000s
5	0m 0.018s	0m 0.002s	0m 0.000s

GCC Compilation Times

Run	Real Time	User Time	Sys Time
1	0m 0.346s	0m 0.013s	0m 0.031s
2	0m 0.342s	0m 0.042s	0m 0.000s
3	0m 0.372s	0m 0.025s	0m 0.024s
4	0m 0.353s	0m 0.013s	0m 0.030s
5	0m 0.349s	0m 0.007s	0m 0.036s

Clang Compilation Times

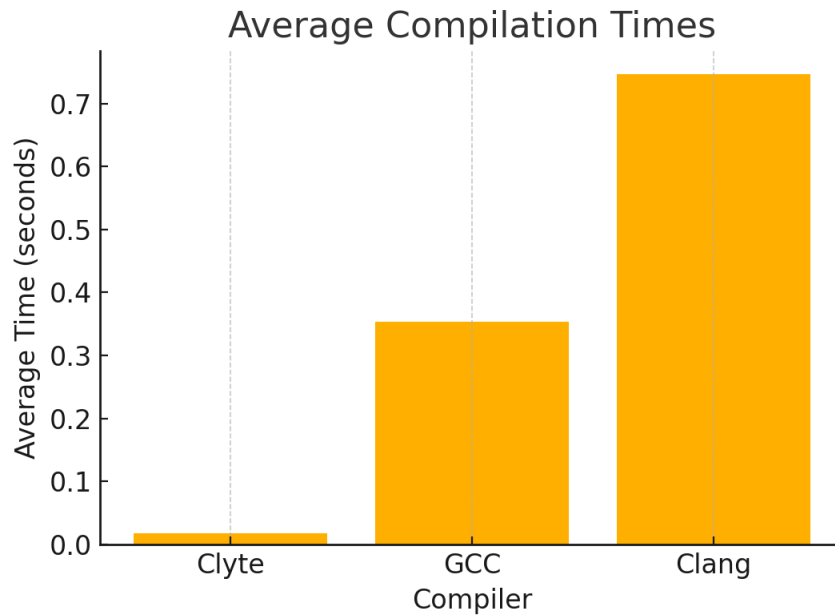
Run	Real Time	User Time	Sys Time
1	0m 0.690s	0m 0.020s	0m 0.060s
2	0m 0.727s	0m 0.021s	0m 0.063s
3	0m 0.739s	0m 0.046s	0m 0.038s
4	0m 0.777s	0m 0.046s	0m 0.041s
5	0m 0.750s	0m 0.040s	0m 0.051s

Average Compilation Times Summary

Compiler	Average Time (seconds)
Clyte	0.018
GCC	0.354
Clang	0.746

Graphical Representation

Bar Chart: Average Compilation Times



- Clyte is by far the fastest, with an average time of 0m 0.018s.
- GCC takes significantly longer at 0m 0.354s on average.
- Clang is the slowest, averaging at 0m 0.746s.

Pie Chart: Time Distribution Across Compilers

Compilation Time Distribution

